# EXPERIMENT NO: 10

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**Division: A** **Batch: A**

**Aim of the Experiment:** Analysis of Cybersecurity Research Papers-Students will explore recent cybersecurity IEEE research papers.

**Learning Objectives:**

- Understand recent research trends in cybersecurity.
- Interpret and analyze technical research papers.
- Explain advanced security techniques and solutions.

**Lab Outcomes:** Student will able to explore recent cybersecurity research papers from IEEE, analyze proposed techniques, and evaluate their effectiveness in solving real-world security problems. It enhances analytical ability, research orientation, and self-learning skills.

**Activity Description:**

Students will independently select a cybersecurity research topic, locate a relevant IEEE research paper, and analyze the methodology, results, and applications. Each student will submit the research findings and provide critical insights.

**Topics for paper selection: Quantum key distribution for secure communication**

## Research Paper Analysis Report

*Title: "Multi-Channel Multi-Protocol Quantum Key Distribution System for Secure Image Transmission in Healthcare" Authors: Bandana Mallick, et al.*

*Source: IEEE Access, Vol. 13, 2025*

### 1. Security Problem Addressed

The research paper addresses the critical challenge of securing electronic health records (EHRs) and highly sensitive medical images, such as X-rays, CT scans, and MRIs, during digital exchange. A primary threat identified is the inefficiency of classical cryptography, as traditional methods like AES and DES rely on treating data as binary sequences, making them too slow to handle the large volumes of highly correlated and redundant data found in real-time medical images. Additionally, the study highlights quantum vulnerabilities where classical techniques based on mathematical algorithms are becoming susceptible to the immense computing power of emerging quantum technologies. Transmission vulnerabilities also pose a significant risk, as medical data remains prone to eavesdropping and tampering when shared between clinics and hospitals over unsecured classical channels. Finally, the authors note that many existing encryption methods result in lossy data revival, which is unacceptable for clinical diagnostic accuracy.

**2. Importance in Modern Cybersecurity**

- Sensitive Data Protection: Healthcare images differ from ordinary data due to strict privacy constraints and their vital role in patient survival and care.
- Scale of Impact: A breach in a centralized healthcare system can expose behavior profiles, location histories, and personal health data of millions of patients simultaneously.
- Regulatory Compliance: With EHRs replacing paper records, institutions face increasing pressure to meet global information preservation and transmission standards.
- Future-Proofing Security: Quantum Key Distribution (QKD) is identified as the next generation of security, exploiting universal laws of quantum mechanics rather than mathematical complexity to ensure privacy.

**3. Techniques / Algorithms Proposed**

The paper proposes a comprehensive framework combining quantum and chaotic defense layers:

Multi-Protocol QKD: Implementation of **BB84** (photon polarization), **CASCADE** (binary search-based error correction), and **Differential-Phase-Shift (DPS)** (phase modulation) to generate and share secret keys.

QBRK Model: A novel "Quantum Bit-plane Representation of the Real Ket" model designed to represent and store images efficiently in quantum states.

Logistic Chaotic System: A mapping sequence used to generate random keys and create a permutation order for pixel positioning.

XOR Operation: A bit-level intensity modification used to scramble the gray-scale values of pixels, ensuring they are unrecognizable to intruders.

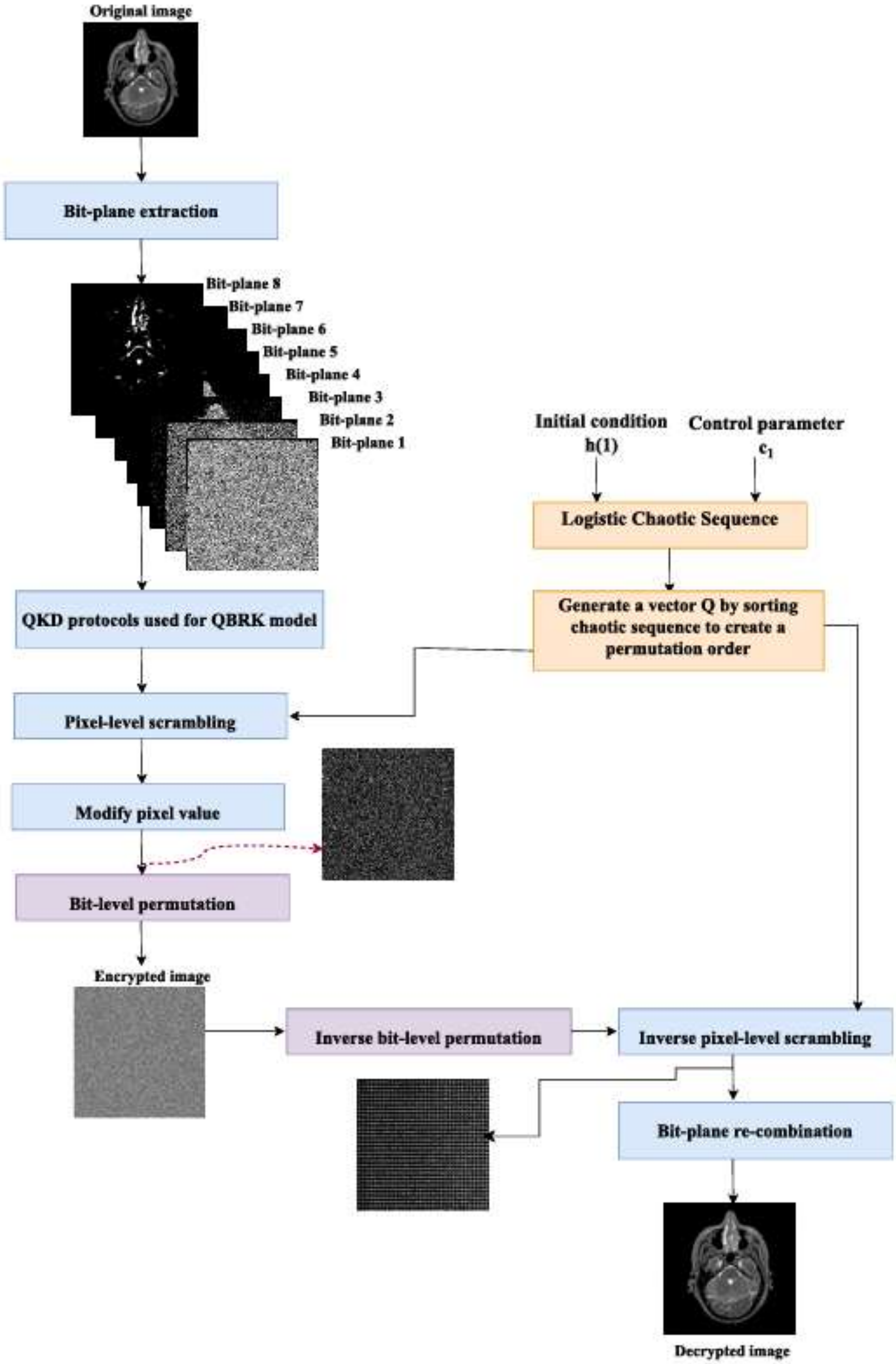


**FIGURE 1.** Flow chart of the proposed algorithm for gray images.

## 4. Architecture / Working Model

The architecture is designed as a comprehensive lifecycle-based data flow where specific security controls are integrated into every stage of the image transmission process. The following table details the operational phases of the proposed Multi-Channel Multi-Protocol QKD system:

| Phase | Description of Activity | Technical Mechanism |
|---|---|---|
| Pre-processing | The medical source image is converted to gray-scale and decomposed into eight distinct bit-planes. | Extraction of 8 bit-planes, prioritizing the top 3 Most Significant Bits (MSBs) which contain approximately 89.17% of the total image information. |
| Scrambling Phase | Entire pixels in the image are repositioned to disrupt the spatial correlation of the data. | Utilization of a logistic chaotic sequence to alter pixel position indices based on a sorted permutation order. |
| Encryption Phase | The bit-level representation of each pixel is modified to ensure maximum confusion. | Application of shared QKD keys (BB84, CASCADE, or DPS) followed by XOR operations and bit-plane shuffling to vary internal bit values. |
| Channel Modeling | The optically modulated signal is prepared for transmission through a physical medium. | RBAC; encrypted data Data is transmitted via a Selector through either an Optical Fiber Channel (OFC) or Free-Space Optics (FSO) at 1550 nm. |
| Decryption | The receiver (Bob) reverses the encryption steps to retrieve the original clinical data. | Execution of inverse bit-level shuffling and inverse pixel-level scrambling using the identical chaotic sequence and reconciled quantum key. |

This structured approach ensures that security is not a single event but a continuous process applied from the initial image slicing to the final reconstruction at the receiver's end.

## 5. Tools and Technologies Used

The experimental framework was implemented on a Core i3 machine with 8 GB of RAM using MATLAB 2017a for encryption and decryption testing. The quantum communication network was designed and simulated using OptiSystem 15.1 software. Key hardware components modeled in the simulation included Mach-Zehnder Modulators (MZM), Continuous-Wave (CW) lasers at 1550 nm, and Avalanche Photo-diodes (APD) for signal detection. The performance of the system was evaluated using 11 different quantitative assessments, including entropy analysis, correlation coefficients, and peak signal-to-noise ratio (PSNR).

## 6. Results Achieved

- High Security Strength: Information entropy reached **7.9896**, which is exceptionally close to the optimal value of 8, signifying that the ciphered image leaks no valuable information.
- Robust Sensitivity: The NPCR and UACI values exceeded **99%** and **33%** respectively, demonstrating that the system is highly resistant to differential attacks.
- Low Error Rates: The BB84 protocol achieved a minimal QBER (Quantum Bit Error Rate) of **0.0028** for FSO channels, outperforming CASCADE and DPS.
- Visual Accuracy: Simulation results confirmed that decrypted medical images maintained high fidelity and were indistinguishable from the original source images.

## 7. Limitations

- Environmental Vulnerability: FSO communication link performance is severely restricted by atmospheric factors such as absorption, scattering, and dense fog, which cause signal decay and link failure over short distances.
- Dark Count Impacts: The Dark Count Rate (DCR) of detectors increases with transmitted power, which compromises the security and accuracy of the protocol, particularly affecting the DPS protocol.
- Computational Overhead: The CASCADE protocol, while efficient for error correction, requires multiple iterations and random permutations, leading to significantly slower encryption times (up to 4.18 seconds compared to BB84's 0.42 seconds).
- Hardware Constraints: Practical deployment is limited by the inability to easily amplify quantum signals without measuring or cloning states, which restricts point-to-point distance.

## 8. Suggested Improvement / Future Research

Proposed Research Direction: Machine Learning-Based Adaptive Protocol Selection. To address the paper's identified research gaps regarding environmental instability, a machine learning model could be developed to monitor channel conditions (like fog density or phase noise) in real-time. The system could then dynamically switch between protocols—selecting **BB84** for high-speed transmission in clear weather and **CASCADE** for robust error correction during turbulent conditions. Furthermore, integrating this with **Post-Quantum Cryptography (PQC)** would create a multi-layer hybrid framework capable of resisting both atmospheric noise and future algorithmic quantum attacks..